Problem Statement

Android Games Data Analysis and Business Insights

Project Objective

This project turns detailed mobile game performance data into insights that game developers and publishers can act on. The goal is to understand what drives higher active-user counts, longer session durations, stronger ratings, and greater revenue across genres, art styles, monetization models, and regions. Excel and Python (Pandas) were used to explore and validate the data, correlation and regression analysis identified the strongest performance drivers, and an interactive three-page Power BI dashboard presents the findings visually.

Problem Statement

The mobile gaming industry generates large volumes of user engagement and revenue data. However, game developers and publishers often struggle to understand which factors contribute to higher active users, longer session durations, better user ratings, and greater revenue.

Specific Gaps Addressed

- It is unclear which genres, art styles, or monetization models generate the most **revenue** versus which simply attract the most **downloads**.
- Developers lack visibility into what keeps players engaged for longer sessions and what brings them back on Day 1, Day 7, and Day 30.
- It is hard to tell which technical and design factors (crash rate, update frequency, app size, multiplayer support) are associated with higher ratings.
- There is no straightforward way to compare regional performance or judge whether marketing spend converts into revenue efficiently.
- Without structured analysis, publishers cannot reliably forecast how revenue and engagement may trend.

Approach

The project cleans and analyzes a 6,142-game dataset, uncovers the patterns that matter, and converts them into insights and forward-looking projections that support better product, marketing, and monetization decisions.

Dataset Overview

Attribute	Details
Dataset	Mobile Games Engagement & Monetization Dataset

Records	6,142 rows (one row per game)
Columns	43
Regions	Global, US, Europe, India, LATAM, SEA
Data Period	2018-2025
Key Columns	genre, sub_genre, monetization_model, art_style, publisher_tier, downloads, active_users, avg_se

Tools Used

Tool	Purpose
Excel	Initial inspection, pivot checks, validation of the raw data
Python (Pandas, NumPy)	Cleaning, aggregation, correlation analysis, EDA
Power BI	Interactive three-page dashboard (revenue, engagement, ratings)

Team Insights Hub - Manisha, Kavya, Nishanth | Teks Academy